FIRENAV: DECISION-ORIENTED RECONSTRUCTION EVIDENCE ROUTING FOR VISION-LANGUAGE NAVIGATION IN INDOOR FIRE ENVIRONMENTS

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Paper under double-blind review

ABSTRACT

In real-world indoor fire environments, visual degradation severely impairs instruction grounding, waypoint prediction, and long-horizon exploration in vision-language navigation (VLN). Although image restoration offers a natural means to mitigate such visual degradation, existing methods primarily optimize reconstruction fidelity, which does not necessarily translate into better navigation decisions and may even corrupt task-critical cues when applied indiscriminately. In this paper, we present FireNav, a decision-oriented reconstruction evidence routing framework that aligns visual restoration with navigation utility in VLN. To advance this task, we first contribute FireVLN-Bench, a paired indoor fire navigation benchmark with clean references preserving task semantics, safety supervision, and matched settings for continuous navigation and robot execution with simulated dynamics. Our FireNav retains native visual tokens and treats latent reconstruction features as auxiliary evidence, where a lightweight router leverages visual features, task context, and prior execution states to determine whether such evidence should be admitted and how strongly it should contribute to the visual tokens. To learn when reconstruction is beneficial, we further introduce decision utility calibration, which compares action prediction losses with and without candidate evidence under the same state, thereby supervising evidence selection according to its contribution to waypoint, heading, and arrival prediction. Experiments demonstrate that FireNav consistently outperforms state-of-the-art baselines in both navigation success and path efficiency across evaluation settings.

1 INTRODUCTION

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2 RELATED WORK

Vision-Language Navigation. VLN maps visual observations conditioned on language to navigation actions and now spans discrete graph navigation, continuous control, open vocabulary search, persistent exploration, and vision-language-action policies. Recently, OmniNav (Xue et al., 2026) unifies instruction goal, object goal, and point goal navigation with frontier exploration through a fast policy for continuous waypoints and a slow planner with memory. DualVLN (Wei et al., 2026) adopts a dual system design that decouples VLM grounding from frequent trajectory execution using explicit pixel goals and implicit latent goals. However, most mainstream VLN methods assume clear or nominal visual observations, leaving robustness under adverse visual conditions less explored. AlldayWalker (Wang et al., 2026c) addresses this problem by adapting a navigator across low light, overexposed, and scattering environments while mitigating forgetting across scenarios. Such adaptation highlights the need for robust visual representations but does not study how reconstruction features can support closed loop navigation without replacing the pretrained visual pathway. In this paper, we explore robust VLN under degradation in indoor fire environments, where compound visual degradations jointly challenge perception and decision making.

Visual Reconstruction. Visual reconstruction aims to recover visibility, structure, and appearance from degraded observations, with most existing methods optimizing image fidelity or perceptual quality. This objective also extends to fire scenes obscured by smoke. Huang et al. (Huang et al., 2021) propose DesmokeGAN, which uses attention-guided adversarial learning for image desmoking and smoke detection, while SmokeBench (Jin et al., 2025) provides aligned real-world smoke/clean surveillance pairs for early-stage fire desmoking. Given the shared scattering and contrast attenuation effects of smoke and haze, recent dehazing methods also offer relevant reconstruction priors. IPC-Dehaze (Fu et al., 2025) iteratively refines discrete visual codes for real-world dehazing, while DCL (Fan et al., 2025) couples atmospheric scattering with temporal brightness consistency via shared depth estimation. Despite improving visibility and perception, these methods largely treat reconstructed images as final outputs. In contrast, embodied VLN requires decision-oriented reconstruction. Thus, we use reconstruction features as optional evidence and evaluate their utility in closed-loop navigation through navigation supervision and task-preservation constraints.

Evidence Routing. Moving from perceptual restoration to decision-oriented reconstruction raises a further question of which recovered features should influence the task representation. UniRestore (Chen et al., 2025) extends restoration toward downstream tasks through task aware feature adaptation and fusion. TaskTok (Lee et al., 2026) selectively refines restoration tokens relevant to the task rather than treating the full latent sequence uniformly. This selective use of latent evidence also appears in embodied decision making. DualVLN (Wei et al., 2026) transfers compact latent goals from a slow VLM planner to a fast trajectory policy. x^2-Fusion (Guo et al., 2026a) adjusts heterogeneous feature contributions according to their reliability under degradation. In indoor fire scenes, reconstruction cues may recover obscured evidence but also distort semantics critical to the task. FireNav therefore introduces decision-guided reconstruction evidence routing to use reconstruction evidence only when it is reliable and beneficial to navigation decisions.

3 Conclusion

We have presented FireNav, a decision-oriented reconstruction evidence routing framework for robust VLN in indoor fire environments. To advance benchmarking research in this field, FireVLN-Bench is first introduced as a paired benchmark with clean references, safety supervision, and matched settings for continuous navigation and robot execution. Our method preserves native visual tokens while selectively incorporating reconstruction evidence through lightweight routing and decision utility calibration. Extensive experiments demonstrate consistent improvements over state-of-the-art baselines in navigation success and path efficiency across evaluation settings.

Vision-language navigation (VLN) requires an embodied agent to ground natural language instructions in visual observations and translate them into navigation behavior. Recent systems improve spatial reasoning by modeling semantics, geometry, agent state, and task progress (Zeng et al., 2026; Guo et al., 2026b). Other designs couple deliberative reasoning with continuous waypoint or trajectory execution (Xue et al., 2026; Wei et al., 2026). Indoor fire environments expose a different robustness challenge because smoke, low illumination, glare, and motion blur corrupt the observations on which both reasoning and control depend.

Visual degradation in embodied agents affects not only perception but also downstream cognition, decision making, and execution (Li et al., 2026a; Wang et al., 2026c). Studies of physically grounded navigation likewise show that observation errors and execution dynamics can jointly degrade task performance (Wang et al., 2025; He et al., 2026). In fire scenes, obscured doorways, people, signs, or obstacle boundaries can cause incorrect waypoints and stopping decisions. Executing such decisions changes subsequent observations and can compound the error. Restoration must therefore recover visibility without suppressing hazards or targets that define the task.

Recent universal restoration methods handle heterogeneous degradations using task or domain prompts and adaptive degradation routing (Dong et al., 2026; Lin et al., 2026). Other work extends reconstruction toward downstream perception or retrieves compact degradation priors for efficient restoration (Chen et al., 2025; Wang et al., 2026a). Yet perceptual relevance does not establish utility for a particular navigation decision. Enhancing an irrelevant region may not improve the next action, reconstructing every observation adds computation, and unconditional feature fusion can disturb reliable native representations. We therefore ask *when reconstruction should influence navigation and how strongly it should modify native visual representations*.

Recent embodied models explore dynamic visual routing and constrained adaptation to specialize perception while preserving pretrained multimodal representations (Wang et al., 2026d; Huang et al., 2026). Related designs pass compact latent or spatial priors to downstream action modules rather than directly modifying the vision-language backbone (Wei et al., 2026; Zhang et al., 2026b). Motivated by this need, we present FireNav, a framework for decision-oriented reconstruction evidence routing in indoor fire VLN. FireNav retains native visual tokens and treats multiscale reconstruction features as auxiliary evidence. Cross attention reads evidence within each view, while a lightweight router uses visual features, task context, and prior execution states to allocate contributions across images and tokens. Bounded residual fusion constrains the resulting representation change. Only the evidence interface and router are trained, while the navigation backbone and reconstruction encoder remain frozen.

Recent navigation methods use relative action quality and semantic task progress to supervise decision making (Li et al., 2026b; Wang et al., 2026b). Related embodied policies also adapt reasoning or execution according to agent state, task progress, or action uncertainty (Guo et al., 2026b; Liang et al., 2026). FireNav instead uses the incremental utility of reconstruction evidence for the current decision. At the same navigation state, we compare predictions with and without candidate reconstruction features from one view. Differences in waypoint, heading, and arrival losses supervise evidence admission relative to the native navigator. These comparisons are performed offline, so inference evaluates only the routed path. Regularization preserves reliable behavior under clean observations, and caching reduces repeated encoding of historical images.

We also introduce FireVLN-Bench, a paired benchmark for indoor fire navigation. As shown in Figure 1, it combines diverse indoor scenes, hazard layouts, and evolving degradations with clean references that preserve fire and task semantics. Paired observations share camera poses and scene states, while matched episodes support continuous navigation and robot execution with simulated dynamics. Shortest paths and trajectories that account for risk enable evaluation of navigation efficiency and safety. Comparisons include reconstructed RGB inputs, adapters with matched capacity, and alternative routing strategies under common execution settings. Experiments show that FireNav outperforms the strongest matched baseline by 4.3/4.9 SR/SPL points in the physical environment and 5.3/6.0 points in the continuous environment. Meanwhile, it limits the clean SR drop to 1.3 and 0.4 points, respectively, with only an 8.4% increase in P95 latency over the native navigator.

Our contributions are summarized below.

- We introduce FireNav, which selectively routes reconstruction features into a frozen navigator for indoor fire VLN while retaining its native visual interface.

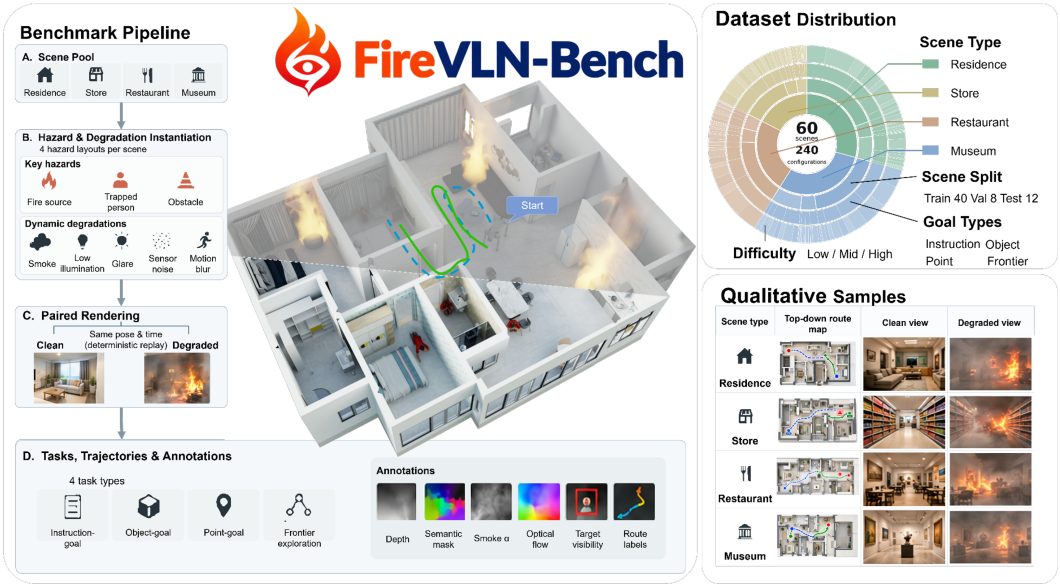


Figure 1: **Overview of FireVLN-Bench.** The figure shows construction pipeline indoor fire scene dataset statistics and qualitative samples.

- We develop decision utility calibration and bounded residual fusion to supervise evidence admission through changes in action prediction and control the resulting representation changes.
- We construct FireVLN-Bench, a paired benchmark with task-preserving clean references and matched continuous and physical navigation settings, enabling controlled evaluation of degradation robustness, safety, reconstruction utility, and execution efficiency.

4 Indoor Fire Vision-Language Navigation Benchmark

Overview. To study robust VLN in indoor fire environments, we construct FireVLN-Bench, a paired benchmark that evaluates the effects of visual degradation and reconstruction on perception and navigation in a closed loop. It covers diverse indoor scenes, hazard layouts, and temporal degradations under matched settings for continuous navigation and robot execution with simulated dynamics. These settings support controlled evaluation of robustness, navigation, safety, and performance recovery through reconstruction. Figure 1 summarizes the benchmark construction pipeline, dataset distribution, and qualitative examples.

Indoor Fire Environment Modeling. We denote an environment configuration by $\mathcal{E}$, comprising scene geometry, materials, collision assets, and a fixed layout of fire sources, trapped persons, and task obstacles. For each camera view, degraded and clean observations at time t are rendered as

$$\begin{aligned} I_t^d &= \mathcal{R}(\mathcal{E}, p_t, \mathcal{D}_t), \\ I_t^c &= \mathcal{R}(\mathcal{E}, p_t, \mathcal{D}_t^c), \end{aligned} \tag{1}$$

where $\mathcal{R}$ denotes rendering and image formation, p_t is the camera pose, and $\mathcal{D}_t$ specifies smoke, illumination, glare, sensor noise, and motion blur. The reference state $\mathcal{D}_t^c$ suppresses nuisance degradations while retaining fire cues and task semantics. Both observations share $\mathcal{E}$, p_t, and the underlying fire and object states. This pairing applies to recorded frames and controlled trajectory replay.

Benchmark Construction. FireVLN-Bench is built from indoor scenes curated from SpatialVerse InteriorAgent and SAGE-3D InteriorGS USDZ (SpatialVerse Research Team, Manycore Tech Inc., 2025; Miao et al., 2026), including residences, convenience stores, restaurants, and museums. We

retain only scenes with valid collision geometry and tasks that are feasible under clean observations. InteriorAgent provides editable mesh scenes, while SAGE-3D combines 3D Gaussian appearance with aligned collision assets.

To simulate fire conditions, we combine each source scene with multiple hazard layouts to form configurations $\mathcal{E}$. Seeded schedules control fire intensity and the evolution of $\mathcal{D}_t$. Using the renderer $\mathcal{R}$ in Eq. equation 1, we generate paired observations (I_t^d, I_t^c) at each recorded pose p_t. The clean rendering state $\mathcal{D}_t^c$ produces the primary reference I_t^c for task preservation. We also provide diagnostic images with visual degradation disabled and annotations including smoke alpha, depth, semantic masks, optical flow, and target visibility. Each configuration is checked for deterministic replay, connectivity, collision consistency, and pair alignment. Episodes include a shortest trajectory and an alternative that accounts for collision risk, fire proximity, dense smoke, blocked routes, and excessive turning.

Task Formulation and Splits. Following continuous and physically embodied VLN settings (Krantz et al., 2020; Wang et al., 2025), FireVLN-Bench contains 60 unique indoor scenes with four hazard layouts each, yielding 240 configurations $\mathcal{E}$. It provides two matched tracks whose canonical episodes share the environment configuration, task, initial state, and seeded degradation schedule. FireVLN-Bench-CE evaluates continuous navigation without teleportation between navigation graph nodes, prior knowledge of the environment topology, or oracle control of local motion. FireVLN-Bench-PE executes the same tasks using simulated robot dynamics and a locomotion controller, introducing contact dynamics, actuation error, camera disturbance, falls, and deadlocks.

The benchmark covers instruction following, object search, point goal navigation, and frontier exploration. For FireNav, the input image set $\mathcal{I}_t$ comprises selected history and current views from the degraded observation stream I_t^d. Each sample contains up to 20 history frames with their recorded poses p_t, current views from the left, front, and right, and five future local waypoints with heading and arrival labels. Corresponding references I_t^c support reconstruction evaluation and comparisons under clean observations. Following protocols for generalization across environments (Ku et al., 2020), the scenes are split into 40 training scenes, 8 unseen scenes for validation, and 12 unseen scenes for testing, corresponding to 160, 32, and 48 hazard configurations. Validation on seen scenes uses separate episodes from the training environments. Splits are stratified by scene category and asset source. All configurations $\mathcal{E}$ derived from the same scene family remain in the same scene partition. Clean and degraded pairs, CE and PE counterparts, and nearby trajectories remain within the same episode split. Episodes involving extended navigation additionally provide rotational views, candidate frontiers, memory proxies, subgoals, and labels for safety and information gain.

5 METHOD

5.1 OVERALL FRAMEWORK

As shown in Figure 2, FireNav augments a pretrained navigator with reconstruction evidence while retaining its native visual pathway. We instantiate the framework on OmniNav (Xue et al., 2026) and preserve its visual encoder, vision-language backbone, waypoint interface, and planning hierarchy. The navigator and reconstruction encoder remain frozen, while the evidence interface and router are learned, limiting adaptation to auxiliary representations (Wei et al., 2026; Huang et al., 2026). At step t, the agent receives a task condition ℓ_t and current and historical images $\mathcal{I}_t = \{I_{t,i}\}$ with pose annotations. Intermediate encoder features support fast execution, and multiscale features support slow planning. The slow system selects subgoals from memory and candidate frontiers. The fast system predicts $H = 5$ planar waypoints with headings and arrival probabilities, denoted by $\widehat{Y}_t$.

Native token interface. Reconstruction features provide an alternative to using restored RGB as the navigation input (Chen et al., 2025; Lee et al., 2026). For image $I_{t,i}$, the native encoder produces $V_{t,i} \in \mathbb{R}^{N_i \times d_v}$. An alignment module projects and pools the reconstruction features into tokens $Z_{t,i}^b \in \mathbb{R}^{M_i^b \times d_a}$ for branch $b \in \{\text{fast}, \text{slow}\}$. Native tokens retrieve evidence from the same image through cross attention (Jiang et al., 2026; Chen et al., 2025),

$$Q_{t,i}^b = V_{t,i} W_Q^b, \qquad C_{t,i}^b = \mathrm{MHA}\left(Q_{t,i}^b, Z_{t,i}^b, Z_{t,i}^b\right). \tag{2}$$

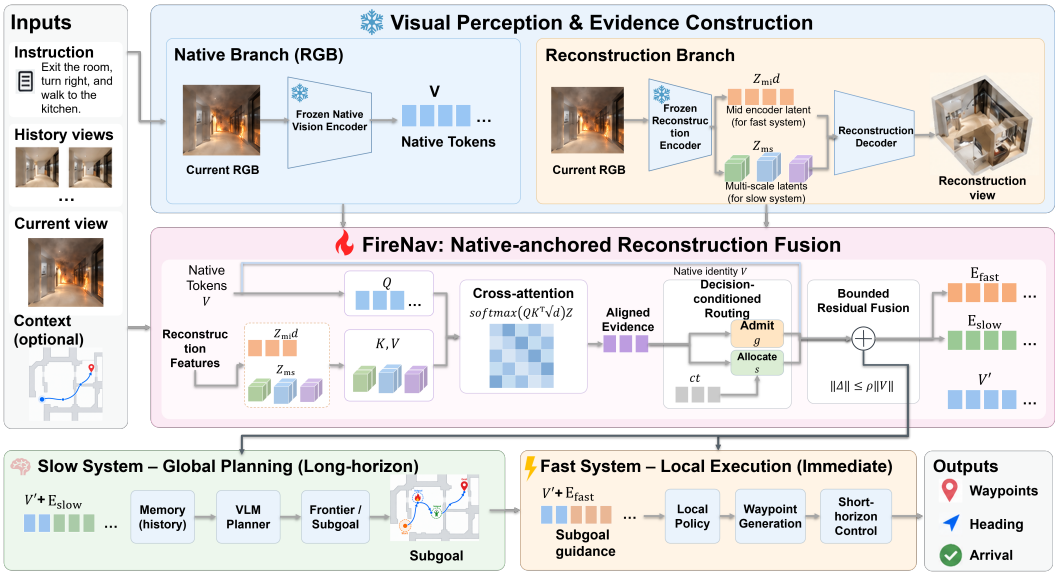


Figure 2: **Overview of FireNav.** A frozen native branch provides visual tokens, while a frozen reconstruction branch exposes mid-level features for fast execution and multiscale features for slow planning. FireNav aligns reconstruction features with native tokens, routes them according to the current decision state, and injects bounded residual evidence while preserving the native visual identity. The slow system uses multiscale evidence for memory-based subgoal planning, whereas the fast system uses mid-level evidence for local waypoint prediction.

The evidence length M_i^b need not equal N_i. The readout occupies the original visual token slots, leaving text, coordinate tokens, and sequence length unchanged. Appendix A.1 details feature selection and token construction.

5.2 Decision-Guided Evidence Routing

Adaptive routing allocates visual information according to input demands and reliability (Wang et al., 2026d; Guo et al., 2026a). FireNav additionally learns whether that information improves the current navigation prediction. It separates evidence admission, spatial allocation, and control of the resulting representation change.

Decision conditioning. Navigation depends on task progress and previous actions as well as current observations (Wang et al., 2026b; Li et al., 2026b). We condition image admission on visual statistics $q_{t,i}$ and context c_t, which includes the task, an available subgoal, the previous waypoint prediction, and measured execution progress. Previous waypoints use the current body frame, and unavailable context is masked. The context excludes the current prediction, clean references, and future outcomes. For each branch, the gates are

$$\begin{aligned} g_{t,i}^b &= \sigma\big(f_g^b([\bar{Q}_{t,i}^b, \bar{C}_{t,i}^b, q_{t,i}, c_t])\big)\,, \\ s_{t,i,j}^b &= \sigma\big(f_s^b([Q_{t,i,j}^b, C_{t,i,j}^b])\big)\,, \end{aligned} \tag{3}$$

where σ denotes sigmoid, bars indicate token averages, brackets denote concatenation, and f_g^b, f_s^b are lightweight MLPs. The scalar $g_{t,i}^b$ sets the contribution of an image, and $s_{t,i,j}^b$ distributes it across native tokens.

Bounded residual evidence. Small gates can still produce large updates when projected features have large magnitudes. To retain the native representation during auxiliary fusion (Jiang et al., 2026;

Huang et al., 2026), we bound the actual residual rather than the gate alone,

$$\Delta^b_{t,i} = \gamma_b g^b_{t,i}\big(s^b_{t,i} \odot (C^b_{t,i} W^b_O)\big), \tag{4}$$

$$E^b_{t,i} = \Delta^b_{t,i} \min\left(1, \frac{\rho_b \|V_{t,i}\|_F}{\|\Delta^b_{t,i}\|_F + \epsilon}\right), \tag{5}$$

$$\widetilde{V}^b_{t,i} = V'_{t,i} + b_{t,i} E^b_{t,i}, \qquad V'_{t,i} \equiv V_{t,i}. \tag{6}$$

Here, γ_b is a learned scale, W^b_O is the output projection, ρ_b bounds relative perturbation, $\epsilon > 0$ stabilizes normalization, and $b_{t,i} \in \{0, 1\}$ indicates valid evidence. Token weights are broadcast over channels. Initializing γ_b to zero recovers the native representation at initialization. Invalid evidence bypasses fusion. This constraint bounds feature changes, not action safety. The intervention ratio and validity checks are given in Appendix A.2.

Decision utility calibration. Decision supervision can use action quality beyond direct imitation (Li et al., 2026b; Wang et al., 2026b). We supervise admission through the change in action prediction caused by candidate reconstruction evidence. The evidence transform and token allocation are first learned with admission fixed open, then frozen. On a disjoint subset of training episodes, we compare the native prediction $\widehat{Y}^0_t$ with $\widehat{Y}^{+i}_t$, obtained by admitting the frozen candidate from one view. Other observations, task inputs, and model states remain unchanged. The utility is

$$u_{t,i} = \mathrm{sg}\Big[\mathcal{L}_{\mathrm{act}}(\widehat{Y}^0_t, Y^\star_t) - \mathcal{L}_{\mathrm{act}}(\widehat{Y}^{+i}_t, Y^\star_t)\Big], \tag{7}$$

where sg stops gradients and $\mathcal{L}_{\mathrm{act}}$ combines position, heading, and arrival errors. Utilities above $\delta > 0$ label beneficial evidence, values below $-\delta$ label harmful evidence, and the neutral interval is excluded. Binary cross entropy trains the corresponding admission gate. The candidate comparison does not depend on that gate, avoiding labels determined by its current admission decision. These comparisons run only offline. They measure isolated view interventions, not additive contributions or guaranteed episode success. The full action loss and calibration procedure appear in Appendix A.3.

5.3 Optimization and Execution

Preservation and task learning. We train on clean and degraded episodes with task learning and representation preservation (Huang et al., 2026; Wei et al., 2026),

$$\mathcal{L} = \mathcal{L}^{(h)}_{\mathrm{nav}} + \lambda_{\mathrm{ret}}\mathcal{L}_{\mathrm{ret}} + \lambda_{\mathrm{int}}\mathcal{L}_{\mathrm{int}} + \lambda_u \mathcal{L}_u. \tag{8}$$

The direct head uses the action loss, and the flow head retains its native flow matching objective. Retention penalizes changes on clean states where the native prediction meets expert tolerances. The intervention loss penalizes actual residual magnitude. Candidate learning omits $\mathcal{L}_u$. Calibration fixes the candidate and optimizes admission with the full objective. Appendix A.4 gives the loss terms and parameter updates.

Cached execution. Following navigation systems that reuse visual history (Xue et al., 2026; Wei et al., 2026), we cache frozen reconstruction features for their original frames and views. Routing is recomputed using the current context. The slow system selects a subgoal with multiscale evidence, and the fast system uses intermediate features for waypoint prediction. Missing, incompatible, or unfinished evidence follows the native path. Caching avoids repeated encoding, not the cost of processing new images. The inherited planning logic and locomotion controller remain unchanged. Cache keys and scheduling details are provided in Appendix A.5.

6 Experiments

6.1 Experimental Settings

Benchmarks and tasks. We evaluate on the matched CE and PE tracks of FireVLN-Bench (Section 4). The main comparison uses instruction following on the unseen test split. Object search, point goal navigation, and frontier exploration are reported in Appendix E. Methods share task specifications, initial states, hazard layouts, degradation schedules, and episode manifests.

Table 1: Quantitative comparison on FireVLN-Bench unseen test split. Among methods with degraded observations, best and second-best results are highlighted in bold and underline, respectively.

Method	Venue	Physical Environment				Continuous Environment			
		NE↓	OS↑	SR↑	SPL↑	NE↓	OS↑	SR↑	SPL↑
Native VLN (clean)	-	4.58	68.2	54.1	42.3	3.32	79.7	67.5	56.6
Native VLN (degraded)	-	7.34	43.0	26.8	17.3	5.94	56.3	41.8	29.9
Reconstructed RGB Frontends									
AdaIR (Cui et al., 2025)	ICLR'25	7.10	45.1	29.4	19.7	5.82	57.2	42.7	30.5
DCPT (Hu et al., 2025)	ICLR'25	6.68	48.8	34.1	25.1	5.33	62.0	47.5	36.6
MoCE-IR-S (Zamfir et al., 2025)	CVPR'25	<u>6.08</u>	50.9	34.8	<u>27.8</u>	5.21	62.9	<u>51.2</u>	38.3
DFPIR (Tian et al., 2025)	CVPR'25	6.65	49.5	32.9	23.7	5.46	60.9	46.2	34.9
R2R-Compos (Wang et al., 2026a)	CVPR'26	6.52	50.5	34.5	24.8	5.32	61.9	48.2	36.6
DATPRL-IR (Dong et al., 2026)	ICLR'26	6.68	49.0	33.5	23.7	5.44	60.8	47.2	35.6
HOGformer (Wu et al., 2026)	AAAI'26	6.84	47.7	31.5	22.3	5.57	60.0	44.5	33.1
BiLaLoRA (Zhang et al., 2026a)	CVPR'26	6.34	<u>54.2</u>	35.6	25.2	<u>4.86</u>	63.1	48.8	<u>40.1</u>
Task Adaptation and Reconstruction Evidence									
Raw RGB + Adapter	-	6.35	52.4	35.3	25.2	5.04	64.6	49.8	38.6
Restored RGB + Adapter	-	6.16	53.9	<u>37.3</u>	27.1	4.99	<u>66.2</u>	50.8	39.5
FireNav	**Ours**	**5.65**	**59.0**	**41.6**	**32.0**	**4.53**	**68.8**	**56.1**	**45.5**

Metrics. We report Navigation Error (NE), Oracle Success (OS), Success Rate (SR), and Success weighted by Path Length (SPL) (Xue et al., 2026; Wei et al., 2026; Wang et al., 2025). PE additionally reports collision, fall, and stuck rates. Local action prediction uses ADE, FDE, heading error, and arrival F1 over five waypoints. Separate clean evaluations measure retention. NE and displacement errors are in meters, heading error is in degrees, and OS, SR, and SPL are percentages.

Baselines. We compare AdaIR (Cui et al., 2025), DCPT (Hu et al., 2025), MoCE-IR-S (Zamfir et al., 2025), DFPIR (Tian et al., 2025), R2R-Compos (Wang et al., 2026a), DATPRL-IR (Dong et al., 2026), HOGformer (Wu et al., 2026), and BiLaLoRA (Zhang et al., 2026a), ordered by publication year. Each frontend supplies reconstructed RGB to the same navigator. Raw RGB + Adapter controls for added capacity and navigation supervision. Restored RGB + Adapter also matches FireNav's reconstruction model. FireNav uses the R2R-Compos encoder and trains only the evidence interface and router. Native navigation with clean or degraded inputs provides reference conditions.

Implementation. Comparisons share the navigation checkpoint, visual history, waypoint interface, controller, and STOP rule. Reconstruction weights remain frozen for each selected encoder. The evidence reader uses width 384 and six attention heads. The main results use the direct waypoint head, with flow variants reported in Appendix G.1. Runtime includes auxiliary encoding and fusion and reports P50/P95 latency, update frequency, and peak memory under the same execution setting.

6.2 Main Results

Table 1 reports 41.6 SR and 32.0 SPL for FireNav in PE, and 56.1 SR and 45.5 SPL in CE. Relative to the native degraded navigator, SR/SPL increase by 14.8/14.7 points in PE and 14.3/15.6 points in CE. Among methods using degraded observations, FireNav ranks first in all eight columns. The best competing SPL comes from MoCE-IR-S in PE and BiLaLoRA in CE, with margins of 4.2 and 5.4 points, respectively.

Against Restored RGB + Adapter, which controls for reconstruction model choice and task adaptation, FireNav gains 4.3/4.9 SR/SPL points in PE and 5.3/6.0 in CE. NE decreases by 0.51 and 0.46 m. These comparisons favor routed features over reconstructed RGB even after task adaptation. Relative to the native clean and degraded conditions, FireNav recovers 54.2%/58.8% of the SR/SPL gap in PE and 55.6%/58.4% in CE. The clean reference remains ahead in both tracks.

Table 2: Ablation study of FireNav. Clean SR measures retention of the native behavior under clean observations. All variants use the same navigation backbone and evaluation protocol.

Variant	Physical Environment			Continuous Environment		
	SR↑	SPL↑	Clean SR↑	SR↑	SPL↑	Clean SR↑
Full FireNav	**41.6**	**32.0**	52.8	**56.1**	**45.5**	67.1
Always-on evidence	36.1	26.0	50.4	50.5	38.6	65.6
w/o decision context	38.9	29.0	53.2	53.7	42.1	67.1
w/o quality statistics	39.8	30.0	53.6	54.4	43.1	66.8
w/o utility calibration	37.8	28.0	52.9	52.2	40.6	66.7
w/o residual control	39.9	29.7	51.6	53.2	41.9	66.2
w/o clean retention	42.8	33.0	48.6	57.2	46.3	63.7
Image gate only	40.8	30.8	53.8	55.4	44.4	67.8
Random encoder + trained interface	37.8	27.5	52.8	51.6	39.7	67.0

Table 3: Evidence intervention analysis with the same trained FireNav checkpoint. Evidence content or router context is modified without retraining the navigator.

Intervention	Physical Environment			Continuous Environment		
	SR↑	SPL↑	Clean SR↑	SR↑	SPL↑	Clean SR↑
Matched evidence	**41.6**	**32.0**	52.8	**56.1**	**45.5**	67.1
Exact bypass	26.8	17.3	54.1	41.8	29.9	67.5
Zero spatial features	28.0	18.0	54.0	41.2	29.5	68.1
Shuffled content	29.3	19.0	53.6	42.4	30.4	66.9
Incorrect view content	27.3	17.4	53.4	40.5	29.2	67.3
Incorrect time content	26.9	17.1	53.3	40.0	28.6	67.2
Shuffled router context only	38.2	28.7	53.8	52.2	41.0	67.4
Invalid metadata / bypass	26.8	17.3	54.1	41.8	29.9	67.5

6.3 Analysis of Decision-Guided Evidence Routing

Component analysis. Table 2 separates evidence selection from preservation. Unconditional fusion reduces PE/CE SPL by 6.0/6.9 points. Removing decision context costs 3.0/3.4 points, and removing utility calibration costs 4.0/4.9 points. Without residual control, SPL drops by 2.3/3.6 points and clean SR also declines. Using only an image gate loses 1.2/1.1 SPL points, indicating a benefit from spatial allocation.

The clean retention comparison shows a tradeoff. Removing it increases degraded SPL by 1.0/0.8 points but lowers clean SR by 4.2/3.4 points. The full model sacrifices a small gain under degradation to retain more of the native performance on clean observations.

Dependence on evidence content. Table 3 changes the inputs to one trained checkpoint. Exact bypass reproduces the native result. Shuffling evidence reduces PE/CE SPL from 32.0/45.5 to 19.0/30.4, and incorrect views or times produce further losses. Shuffling only router context reduces SPL by 3.3/4.5 points. The additional branch is present in these interventions, but its benefit depends on the supplied content and context. The random encoder and matched adapter controls provide complementary tests of reconstruction pretraining and model capacity.

6.4 Action Quality and Efficiency

Action prediction. Relative to Restored RGB + Adapter, FireNav lowers degraded ADE/FDE from 0.555/0.926 to 0.504/0.861 m and heading error from 17.0° to 15.2° (Table 4). Arrival F1 rises from 0.586 to 0.667. Clean ADE remains close to the native value, at 0.506 versus 0.504 m.

Table 4: Action prediction under degraded and clean observations. ADE/FDE are measured in meters, heading error in degrees, and arrival F1 as a fraction.

Method	Degraded Observations				Clean Observations		
	ADE↓	FDE↓	Heading↓	Arrival F1↑	ADE↓	Heading↓	Arrival F1↑
Native VLN	0.666	1.101	21.0	0.474	**0.504**	**17.0**	**0.670**
MoCE-IR-S	0.595	0.991	18.8	0.546	0.516	17.5	0.657
R2R-Compos	0.615	1.021	19.5	0.523	0.522	17.8	0.644
HOGformer	0.615	1.036	19.8	0.509	0.511	17.3	0.663
BiLaLoRA	0.600	1.001	18.6	0.552	0.514	17.4	0.663
Raw RGB + Adapter	0.565	0.956	17.4	0.575	0.507	17.2	0.667
Restored RGB + Adapter	0.555	0.926	17.0	0.586	0.509	17.4	0.663
FireNav	**0.504**	**0.861**	**15.2**	**0.667**	0.506	17.1	0.669

Table 5: End-to-end runtime comparison under the same execution setting. P50/P95 denote update latency.

Method	Additional Cost		End-to-End Latency		System	
	Encoder/Restoration (ms)↓	Fusion (ms)↓	P50 (ms)↓	P95 (ms)↓	Hz↑	Peak GB↓
Native VLN	0.0	0.0	**280.1**	**359.2**	**3.53**	**18.06**
AdaIR	209.9	2.0	491.9	597.4	2.02	20.26
DCPT	96.5	2.0	378.8	464.8	2.62	19.16
MoCE-IR-S	126.1	2.0	408.5	498.1	2.43	19.36
DFPIR	113.9	2.0	396.9	485.0	2.51	19.66
R2R-Compos	78.3	2.0	361.2	445.0	2.75	19.06
DATPRL-IR	134.8	2.0	417.3	508.5	2.38	19.86
HOGformer	270.4	2.0	551.3	668.4	1.80	20.56
BiLaLoRA	71.8	2.0	354.0	436.5	2.80	19.56
Raw RGB + Adapter	0.0	8.0	288.0	367.3	3.43	18.46
Restored RGB + Adapter	77.9	10.0	368.7	452.4	2.69	19.46
FireNav	17.9	12.0	310.1	389.4	3.19	19.16

Runtime. Table 5 reports 17.9 ms for feature encoding and 12.0 ms for fusion. FireNav has a P95 latency of 389.4 ms, 8.4% above the native navigator, and an update frequency of 3.19 rather than 3.53 Hz. Compared with Restored RGB + Adapter, P95 is 63.0 ms lower and frequency increases from 2.69 to 3.19 Hz. Direct use of encoder features avoids RGB decoding on the fast path. Appendix G examines history reuse and prediction heads, and Appendix D reports physical failures.

AI USE STATEMENT

Generative AI tools were used under the authors' direction to assist with literature search, experimental design, software development, result interpretation, figure drafting, and manuscript writing and editing. All AI-assisted code, analyses, figures, citations, and text were reviewed and verified by the authors. The authors take full responsibility for the final content of this work.

ETHICS STATEMENT

FireNav and FireVLN-Bench are intended for research on robust navigation in simulated indoor fire environments. The benchmark contains no human-subject data or personally identifiable information. Because simulation cannot fully capture real fire conditions, the reported performance should not be treated as evidence of safety for real rescue deployment. Real-world use requires additional validation, redundant sensing, and appropriate safety procedures.

REPRODUCIBILITY STATEMENT

Code, configuration files, benchmark generation scripts, data splits, and evaluation tools will be released with the paper. The main text and appendix provide the benchmark construction, model architecture, training objectives, implementation details, evaluation protocol, and ablation settings required to reproduce the reported experiments. All reported tables and figures will be generated from stored experimental outputs using the released evaluation scripts.

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A Additional Method Details

A.1 Evidence Sources and Token Construction

For each observation, the frozen reconstruction encoder provides an intermediate feature $R_{t,i}^{\mathrm{mid}}$ and features $\{R_{t,i}^{l}\}_{l\in\mathcal{L}}$ at multiple scales. The source used by branch b is

$$\mathcal{R}_{t,i}^{b} = \begin{cases} R_{t,i}^{\mathrm{mid}}, & b = \mathrm{fast}, \\ \{R_{t,i}^{l}\}_{l\in\mathcal{L}}, & b = \mathrm{slow}. \end{cases} \tag{9}$$

The alignment module applies projections and spatial pooling for the selected scales, conditioned on image statistics and view identity,

$$Z_{t,i}^{b} = A_{\phi}^{b}\left(\mathcal{R}_{t,i}^{b}, q_{t,i}, \nu_i\right) \in \mathbb{R}^{M_i^b \times d_a}. \tag{10}$$

The statistics $q_{t,i}$ contain brightness, contrast, saturation, and gradient information. The identifier ν_i distinguishes views. Cross attention is restricted to the corresponding image, so each readout has N_i tokens regardless of the evidence length M_i^b. Its projection returns to the native width d_v. Text tokens, coordinate tokens, and multimodal sequence length are unchanged. Reconstruction features are read from the encoder, and the online fast path does not require RGB decoding.

A.2 Residual Validity and Initialization

The validity indicator $b_{t,i}$ is distinct from the branch index b. Frame identity, preprocessing, checkpoint consistency, and finite tensor values are checked before fusion. Invalid packets return the native tokens directly, rather than multiplying invalid values by zero. In Figure 2, $V'_{t,i} \equiv V_{t,i}$ denotes the retained identity stream, not an already fused representation. Valid branch inputs are $V'_{t,i} + E^b_{t,i}$, with evidence injected once.

The learnable residual scale γ_b starts at zero, making the initial interface identical to the native representation. For $\rho_b \geq 0$ and $\epsilon > 0$, the effective intervention is

$$r_{t,i}^{b} = \frac{\|b_{t,i} E_{t,i}^{b}\|_F}{\|V_{t,i}\|_F + \epsilon} \leq \rho_b. \tag{11}$$

This quantity measures the actual update rather than its gate alone. The bound follows from the scaling in Eq. equation 5. It limits representation changes without guaranteeing semantic preservation or safe actions.

A.3 Action Loss and Utility Calibration

The action objective evaluates all H decoded waypoints. For predicted positions $\hat{p}_j$, headings $\hat{\theta}_j$, and arrival probabilities $\hat{a}_j$, with corresponding targets marked by $\star$, it is

$$\begin{aligned}\mathcal{L}_{\mathrm{act}}(\widehat{Y}, Y^{\star}) = \frac{1}{H}\sum_{j=1}^{H}\Big[&\lambda_p \,\mathrm{SmoothL1}(\hat{p}_j - p_j^{\star}) \\ &+ \lambda_\theta\big(1-\cos(\hat{\theta}_j - \theta_j^{\star})\big) \\ &+ \lambda_a \,\mathrm{BCE}(\hat{a}_j, a_j^{\star})\Big].\end{aligned} \tag{12}$$

Coordinates follow the native head's decoding and normalization convention. The utility in Eq. equation 7 compares the native prediction with a frozen candidate admitted from one view. Other observations, task inputs, and model states are held fixed. Utilities within $[-\delta, \delta]$ are excluded from admission supervision,

$$\mathcal{L}_u = \mathbb{E}_{|u_{t,i}|>\delta}\,\mathrm{BCE}(g_{t,i}, \mathbf{1}[u_{t,i} > \delta])\,. \tag{13}$$

Here, $g_{t,i}$ is the admission score associated with the candidate used in the calibrated prediction, with the branch index suppressed. Candidate predictions are computed with admission open, independently of this learned gate. For the flow head, both predictions use the same initial noise and are scored after decoding. Calibration episodes are drawn from the training partition and excluded from candidate learning. The labels concern isolated views and do not assume additive utility. The navigation objective accounts for predictions made with multiple views admitted. No paired comparison is required online.

A.4 Loss Terms and Training Stages

The direct head uses Eq. equation 12. The flow head retains its native objective in normalized action space (Xue et al., 2026; Wei et al., 2026). Their decoding conventions and checkpoints remain separate. The retention term applies only to clean states where native predictions meet expert tolerances. It compares assisted predictions with detached native outputs using position, circular heading, and arrival probability differences. Residual regularization averages the squared intervention ratio,

$$\mathcal{L}_{\mathrm{int}} = \mathbb{E}_t\left[\frac{1}{|\mathcal{B}_t|\,|\mathcal{I}_t|}\sum_{b\in\mathcal{B}_t}\sum_{i}(r_{t,i}^{b})^2\right], \tag{14}$$

where $\mathcal{B}_t \subseteq \{\mathrm{fast}, \mathrm{slow}\}$ contains the active branches.

Candidate learning fixes image admission open and trains the evidence transform and token allocation without $\mathcal{L}_u$. Calibration freezes that candidate and trains admission using Eq. equation 8. Gradients pass through the frozen navigator to the currently trainable interface modules, but do not update its weights. Clean and degraded episodes contribute to the objective.

A.5 Cache Consistency and Scheduling

Cache keys include frame identity, view, preprocessing transform, checkpoint, and numerical precision. Only frozen reconstruction features are reused. Evidence admission is recomputed from the current decision context, rather than reusing a previous fused representation. Historical features remain attached to their original observations and are never substituted for another view or timestamp.

At each fusion boundary, missing, incompatible, or unfinished evidence follows the native path. A pending feature may later be used for its matching frame in visual history. The slow system selects a subgoal using memory and multiscale evidence. The fast system combines that guidance with intermediate features for waypoint prediction. Encoder scheduling and cache reuse do not change the inherited planning logic or locomotion controller. The execution analysis separates new feature extraction, reuse, fusion, and total update latency.

B Additional Experimental Details

B.1 Benchmark Allocation

Table 6: Data allocation in FireVLN-Bench. Validation seen uses held-out episodes from the training scenes and therefore does not introduce additional scene identities.

Split	Scenes	Layouts	Routes/Layout/Task	Episodes/Task	Four-Task Episodes
Train	40	160	100	16,000	64,000
Validation Seen	40	160	25	4,000	16,000
Validation Unseen	8	32	25	800	3,200
Test Unseen	12	48	25	1,200	4,800

Table 6 assigns the 60 scenes to 40 training, 8 unseen validation, and 12 unseen test scenes, each with four hazard layouts. Validation seen reserves episodes from the training scenes without adding scene identities. The main comparison uses 1,200 instruction following episodes per track. CE and PE share canonical tasks but are evaluated separately.

B.2 Evaluation Protocol

Methods share episode manifests, task inputs, degradation schedules, and success definitions. Clean and degraded frames are paired at the same pose for supervision and controlled comparisons. Independent navigation policies can follow different trajectories during closed loop execution. Object search, point goal navigation, and frontier exploration use their respective task criteria. Safety rates include all evaluated PE episodes. Local action errors are computed over five predicted waypoints separately from rollout metrics.

C Extended Routing Analysis

C.1 Encoder and Routing Strategy

Table 7 compares routing strategies within each encoder under the same evidence budget and optimization recipe. FireNav routing gives higher SPL than unconditional fusion and quality gating for all eight encoders. For R2R-Compos, unconditional fusion yields 26.0/38.6 PE/CE SPL, compared with 32.0/45.5 for FireNav. DCPT reaches 32.5/46.0, showing that the selected R2R encoder is not the strongest on every metric.

C.2 Image Quality and Action Utility

The rankings in Table 8 differ across image quality and action error. BiLaLoRA has the highest PSNR and SSIM among restored RGB frontends, but MoCE-IR-S has the lowest degraded ADE. R2R-Compos reduces ADE relative to raw input despite lower PSNR and SSIM. Image quality alone does not capture these changes in navigation prediction.

The image-free prior in Table 9 lowers ADE from 0.666 to 0.620 m but raises heading error from 21.0° to 26.0° and lowers arrival F1 from 0.474 to 0.440. Displacement accuracy can improve without a corresponding gain in orientation or termination prediction. FireNav has lower degraded displacement and heading errors and higher arrival F1 than both this prior and the matched adapters.

D Physical Safety and Termination

Among methods using degraded observations, FireNav has the lowest collision, stuck, and false STOP rates in Table 10. Relative to Restored RGB + Adapter, PE collision decreases from 20.6% to 16.9% and stuck rate from 11.7% to 9.3%. Fall rate is 2.2%, slightly above the adapter's 2.1%, so the advantage does not extend to every physical failure category. CE collision and false STOP rates decrease from 9.9%/7.8% to 7.2%/5.7%.

Table 7: Factorial comparison of reconstruction encoders and evidence routing strategies. Values are SPL (%). Each encoder uses the same evidence budget and optimization recipe.

Encoder	Physical Environment			Continuous Environment		
	Always-on	Quality Gate	FireNav	Always-on	Quality Gate	FireNav
AdaIR	23.6	26.0	**27.8**	35.6	38.0	**40.3**
DCPT	27.0	28.6	**32.5**	39.2	41.2	**46.0**
MoCE-IR-S	26.4	29.4	**32.3**	39.1	42.1	**45.7**
DFPIR	25.4	28.6	**30.8**	37.9	41.2	**44.2**
R2R-Compos	26.0	29.2	**32.0**	38.6	41.9	**45.5**
DATPRL-IR	26.0	28.9	**31.7**	38.3	41.9	**45.4**
HOGformer	25.4	27.8	**30.2**	37.0	40.3	**43.9**
BiLaLoRA	26.7	28.5	**31.3**	38.2	41.4	**44.9**

Table 8: Relationship between image reconstruction quality, action prediction, and RGB reconstruction cost.

Method	Image Quality		Action ADE		Cost
	PSNR↑	SSIM↑	Degraded↓	Clean↓	RGB ms↓
Raw degraded	16.8	0.680	0.666	**0.504**	0.0
AdaIR	15.9	0.648	0.645	0.533	69.9
DCPT	17.8	0.717	0.603	0.518	31.8
MoCE-IR-S	18.2	0.728	**0.595**	0.516	42.0
DFPIR	17.6	0.711	0.608	0.520	38.1
R2R-Compos	16.2	0.658	0.615	0.522	26.1
DATPRL-IR	17.5	0.707	0.610	0.519	44.9
HOGformer	16.6	0.697	0.615	0.511	90.4
BiLaLoRA	**18.6**	**0.745**	0.600	0.514	24.0

Table 11 separates valid termination, goal visitation without success, and failure to visit the goal. Relative to Restored RGB + Adapter, FireNav records 51 more valid stops in PE and 63 more in CE. The PE count of visited but failed episodes also increases, from 199 to 209. Goal discovery and successful termination remain distinct outcomes, motivating separate OS and SR reporting.

E Generalization Across Splits and Tasks

E.1 Seen and Unseen Environments

FireNav exceeds Restored RGB + Adapter on each split in Table 12. Its PE SPL is 41.7 on seen validation and 32.0 on unseen testing, with corresponding CE values of 52.5 and 45.5. The advantage over the adapter persists, but performance still declines across the seen and unseen scene sets.

E.2 Goal-Directed Tasks

Table 13 extends the comparison beyond instruction following. FireNav gains 5.5/5.2 PE/CE SPL points over Restored RGB + Adapter on object search and 4.9/5.4 points on point goal navigation. Point goal scores are higher than object search scores for every listed system.

E.3 Frontier Exploration

Relative to Restored RGB + Adapter, FireNav increases discovery by 5.4 points in PE and 5.7 points in CE (Table 14). The reported mean discovery steps decrease from 303.2/275.5 to 274.6/251.0, and

Table 9: Image-free and capacity controls for action prediction.

Method	Degraded Observations				Clean
	ADE↓	FDE↓	Heading↓	F1↑	ADE↓
Native VLN	0.666	1.101	21.0	0.474	**0.504**
Image-free horizon prior	0.620	1.021	26.0	0.440	0.565
Raw RGB + Adapter	0.565	0.956	17.4	0.575	0.507
Restored RGB + Adapter	0.555	0.926	17.0	0.586	0.509
FireNav	**0.504**	**0.861**	**15.2**	**0.667**	0.506

Table 10: Safety and terminal behavior on FireVLN-Bench. All values are percentages of test episodes.

Method	Physical Environment				Continuous Environment	
	Collision↓	Fall↓	Stuck↓	False STOP↓	Collision↓	False STOP↓
Native VLN (clean)	10.3	1.9	6.3	3.2	3.5	3.3
Native VLN (degraded)	33.2	3.2	16.2	13.5	19.8	14.5
AdaIR	31.0	3.2	15.7	12.7	18.1	13.2
DCPT	23.6	2.2	13.7	10.0	13.2	10.2
MoCE-IR-S	22.5	2.1	12.8	9.0	11.8	10.1
DFPIR	24.8	**1.4**	13.8	9.8	13.8	10.8
R2R-Compos	24.8	2.7	13.7	9.8	13.8	10.5
DATPRL-IR	25.3	2.8	13.9	10.6	14.1	11.3
HOGformer	27.2	2.8	14.5	11.3	15.8	11.8
BiLaLoRA	23.2	2.7	12.8	9.3	12.9	9.5
Raw RGB + Adapter	21.8	2.1	12.3	7.7	11.0	9.2
Restored RGB + Adapter	20.6	2.1	11.7	7.2	9.9	7.8
FireNav	**16.9**	2.2	**9.3**	**5.0**	**7.2**	**5.7**

repetition falls from 35.1%/31.9% to 29.0%/26.0%. These aggregate measures complement the goal reaching results.

F Robustness to Degradation Severity

All listed systems lose performance as degradation increases (Table 15). FireNav retains an advantage at each level. Under severe degradation, its PE/CE SPL is 21.7/31.5, compared with 17.3/25.7 for Restored RGB + Adapter. Absolute performance remains below the mild setting, where FireNav achieves 46.5/56.1 SPL.

G History, Caching, and Prediction Heads

Table 16 shows diminishing gains as history increases. Moving from 10 to 20 frames adds 0.5/0.3 PE/CE SPL points. Recomputing all 23 images leaves the listed navigation scores unchanged but raises P95 from 389.4 to 520.4 ms. Using evidence only from current views reduces P95 to 384.8 ms while losing 2.5/1.3 SPL points. The cache comparison measures reuse of historical features, not avoidance of encoding new observations.

G.1 Prediction Heads

FireNav improves each corresponding native head in Table 17. With FireNav, moving from O-Direct to O-Flow-5 adds 2.3/2.4 PE/CE SPL points and 25.5 ms of P95 latency. O-Flow-10 adds a further 0.3/0.5 points and 39.0 ms over O-Flow-5. Comparisons within each head separate evidence routing from changes in trajectory generation and solver cost.

Table 11: Arrival-state accounting. The first three outcome columns form a disjoint partition of all episodes; False STOP is reported separately.

Setting	N	Valid STOP↑	Visited but Failed↓	Never Visited↓	False STOP↓
Native VLN / PE	1200	321	195	684	13.5
Native VLN / CE	1200	501	175	524	14.5
Restored RGB + Adapter / PE	1200	448	199	553	7.2
Restored RGB + Adapter / CE	1200	610	168	422	7.8
FireNav / PE	1200	**499**	209	**492**	**5.0**
FireNav / CE	1200	**673**	**153**	**374**	**5.7**

Table 12: Generalization across seen validation, unseen validation, and unseen test environments.

Method / Split	Physical Environment		Continuous Environment	
	SR↑	SPL↑	SR↑	SPL↑
Native VLN / Seen Val.	36.4	26.2	45.8	35.0
Native VLN / Unseen Val.	33.5	22.6	40.0	29.5
Native VLN / Unseen Test	26.8	17.3	41.8	29.9
BiLaLoRA / Seen Val.	44.7	34.0	53.8	43.0
BiLaLoRA / Unseen Val.	40.6	29.6	48.0	37.4
BiLaLoRA / Unseen Test	35.6	25.2	48.8	40.1
Restored RGB + Adapter / Seen Val.	47.0	36.4	56.5	45.6
Restored RGB + Adapter / Unseen Val.	42.4	31.4	50.5	39.8
Restored RGB + Adapter / Unseen Test	37.3	27.1	50.8	39.5
FireNav / Seen Val.	**51.8**	**41.7**	**63.0**	**52.5**
FireNav / Unseen Val.	**46.9**	**36.6**	**56.6**	**46.3**
FireNav / Unseen Test	**41.6**	**32.0**	**56.1**	**45.5**

H Cross-System Context

Table 18 provides context across navigation systems. DualVLN differs from the OmniNav-based configurations in backbone, training, and action generation. In the listed comparison, FireNav exceeds DualVLN + R2R RGB by 1.0/8.3 PE/CE SPL points. This cross-system margin cannot isolate the effect of evidence routing. The main comparison holds the navigation backbone fixed for that purpose.

Table 13: Generalization across instruction, object, and point goal tasks.

Method / Task	Physical Environment		Continuous Environment	
	SR↑	SPL↑	SR↑	SPL↑
Native VLN / Instruction	26.8	17.3	41.8	29.9
Native VLN / Object	20.0	12.0	28.1	18.4
Native VLN / Point	48.3	35.8	63.0	50.7
BiLaLoRA / Instruction	35.6	25.2	48.8	40.1
BiLaLoRA / Object	25.3	16.1	34.3	23.5
BiLaLoRA / Point	56.4	42.8	70.1	56.9
Restored RGB + Adapter / Instruction	37.3	27.1	50.8	39.5
Restored RGB + Adapter / Object	27.5	17.8	36.8	24.9
Restored RGB + Adapter / Point	59.2	45.5	71.2	58.2
FireNav / Instruction	**41.6**	**32.0**	**56.1**	**45.5**
FireNav / Object	**33.1**	**23.3**	**40.8**	**30.1**
FireNav / Point	**63.0**	**50.4**	**74.5**	**63.6**

Table 14: Frontier exploration without a prescribed final goal. Discovery and repetition are percentages; Steps denotes the mean number of steps required for discovery.

Method	Physical Environment			Continuous Environment		
	Discover↑	Steps↓	Repeat↓	Discover↑	Steps↓	Repeat↓
Native VLN	45.6	345.2	43.0	53.0	315.9	38.9
BiLaLoRA	52.5	312.0	36.4	58.3	288.4	33.5
Restored RGB + Adapter	54.2	303.2	35.1	61.2	275.5	31.9
FireNav	**59.6**	**274.6**	**29.0**	**66.9**	**251.0**	**26.0**

Table 15: Performance under mild, moderate, and severe visual degradation.

Severity	Method	Physical Environment		Continuous Environment	
		SR↑	SPL↑	SR↑	SPL↑
Mild	Native VLN	44.3	29.3	55.7	39.3
Mild	BiLaLoRA	53.3	38.5	64.3	48.5
Mild	Restored RGB + Adapter	56.0	41.1	67.0	51.4
Mild	**FireNav**	**60.0**	**46.5**	**70.0**	**56.1**
Moderate	Native VLN	24.2	15.5	42.8	31.0
Moderate	BiLaLoRA	33.7	23.7	50.3	38.8
Moderate	Restored RGB + Adapter	34.7	24.9	51.7	40.4
Moderate	**FireNav**	**39.0**	**29.9**	**57.8**	**47.1**
Severe	Native VLN	14.3	9.0	25.7	18.4
Severe	BiLaLoRA	21.7	15.1	30.0	22.9
Severe	Restored RGB + Adapter	24.0	17.3	33.0	25.7
Severe	**FireNav**	**28.3**	**21.7**	**38.7**	**31.5**

Table 16: Effect of visual history and feature caching.

Configuration	Cache		Runtime		PE		CE	
	Calls	Hit (%)	P50	P95	SR	SPL	SR	SPL
No history, 3 current views	3	0.0	206.1	256.5	37.9	28.0	52.5	41.5
4 history + 3 current views	3	57.1	242.1	302.2	40.8	30.7	54.8	43.9
10 history + 3 current views	3	76.9	277.9	348.9	41.2	31.5	56.0	45.2
20 history + 3 current views	3	87.0	310.1	389.4	**41.6**	**32.0**	**56.1**	**45.5**
20 history, front-view evidence only	1	95.2	292.1	369.5	39.2	29.2	54.1	42.8
20 history, current evidence only	3	0.0	306.5	384.8	39.6	29.5	55.2	44.2
20 history, recompute all 23 images	23	0.0	430.3	520.4	**41.6**	**32.0**	**56.1**	**45.5**

Table 17: Comparison of direct and flow based waypoint prediction heads.

Configuration	Physical Environment		Continuous Environment		Latency	
	SR↑	SPL↑	SR↑	SPL↑	P50↓	P95↓
O-Direct / Native	26.8	17.3	41.8	29.9	280.1	359.2
O-Direct / FireNav	41.6	32.0	56.1	45.5	310.1	389.4
O-Flow-5 / Native	30.1	20.6	44.2	33.0	300.1	384.8
O-Flow-5 / FireNav	43.6	34.3	58.6	47.9	329.8	414.9
O-Flow-10 / Native	30.0	20.5	44.6	33.3	330.1	423.3
O-Flow-10 / FireNav	**44.0**	**34.6**	**59.3**	**48.4**	360.0	453.9

Table 18: Cross-system comparison on FireVLN-Bench. This table provides system context only because the compared models differ in backbone, training, action head, and execution budget.

System	Physical Environment		Continuous Environment	
	SR↑	SPL↑	SR↑	SPL↑
OmniNav O-Direct	26.8	17.3	41.8	29.9
DualVLN	38.2	27.5	45.6	34.6
DualVLN + R2R RGB	41.2	31.0	48.6	37.2
FireNav	**41.6**	**32.0**	**56.1**	**45.5**